

Microprocessors and Microcontrollers Lab

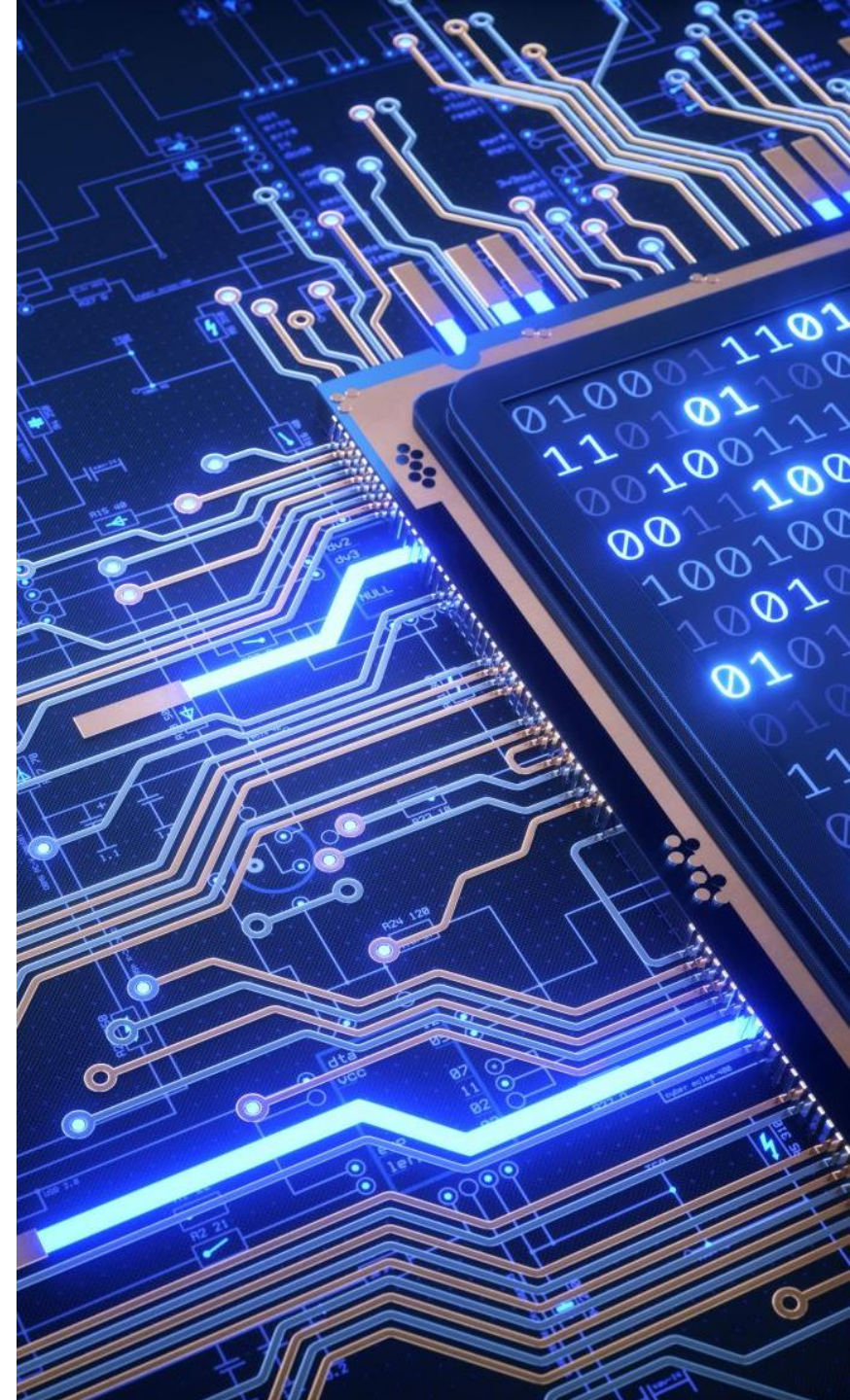
BECE204P

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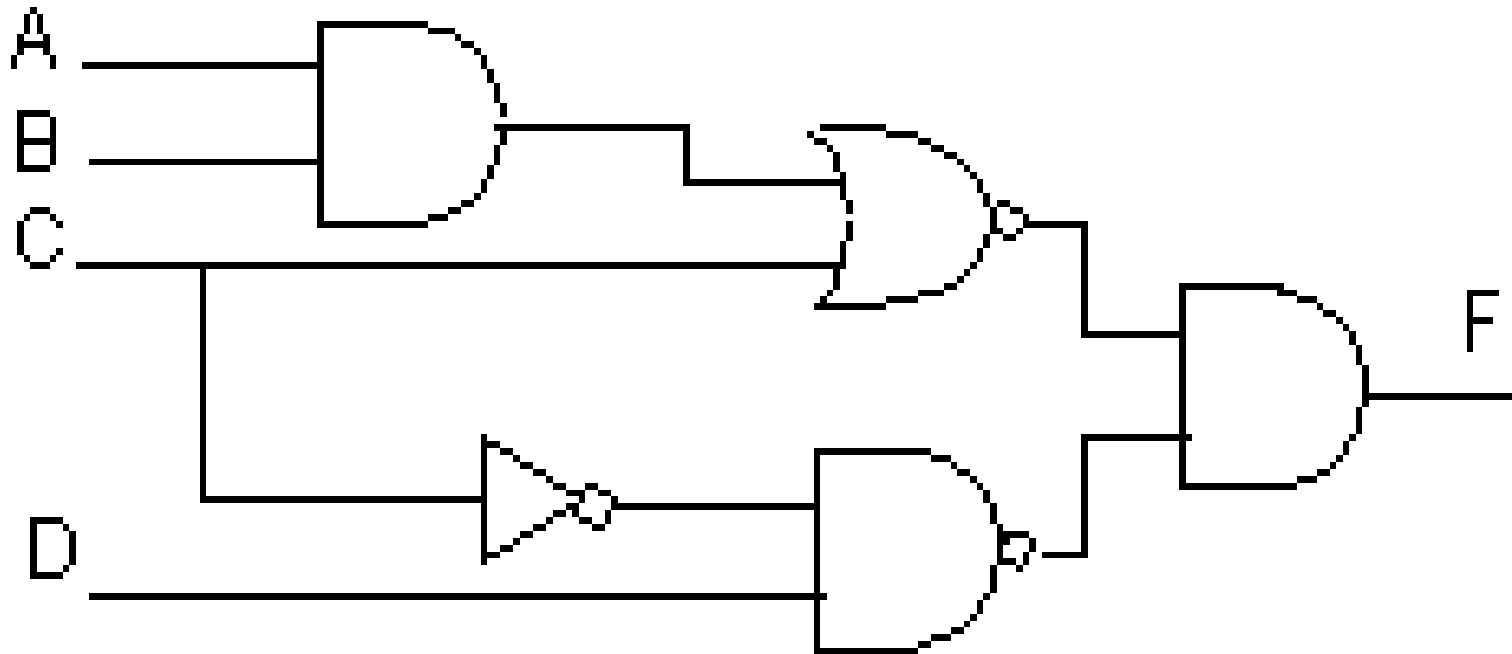


Logical Operations Programming



Assembly Language Programming

Question : To write an assembly code for 8051 Microcontroller for implementation of the given digital circuit using KEIL development tool.

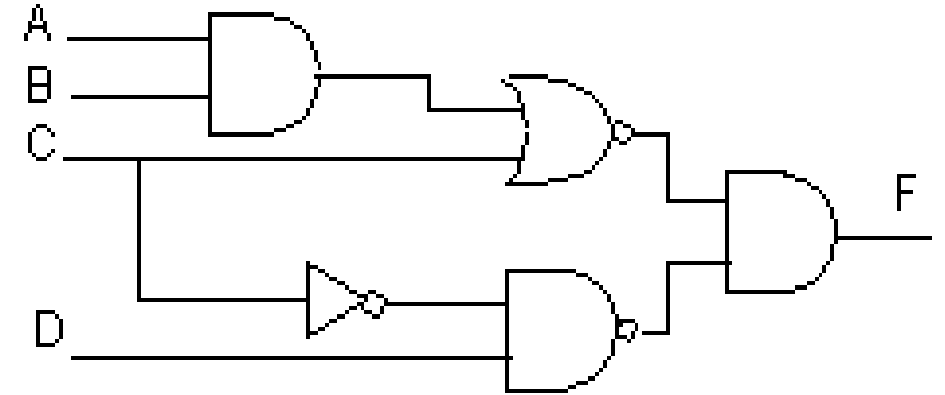


Assembly Language Programming

Solution:

```
ORG 0000H  
SETB ACC.0      ; input A  
SETB ACC.1      ; input B  
CLR ACC.2       ; input C  
SETB ACC.3      ; input D  
MOV C, ACC.0  
ANL C, ACC.1  
ORL C, ACC.2  
CPL C
```

```
MOV ACC.7, C  
MOV C, ACC.2  
CPL C  
ANL C, ACC.3  
CPL C  
ANL C, ACC.7      ; Final output (F)  
HALT: SJMP HALT  
END
```



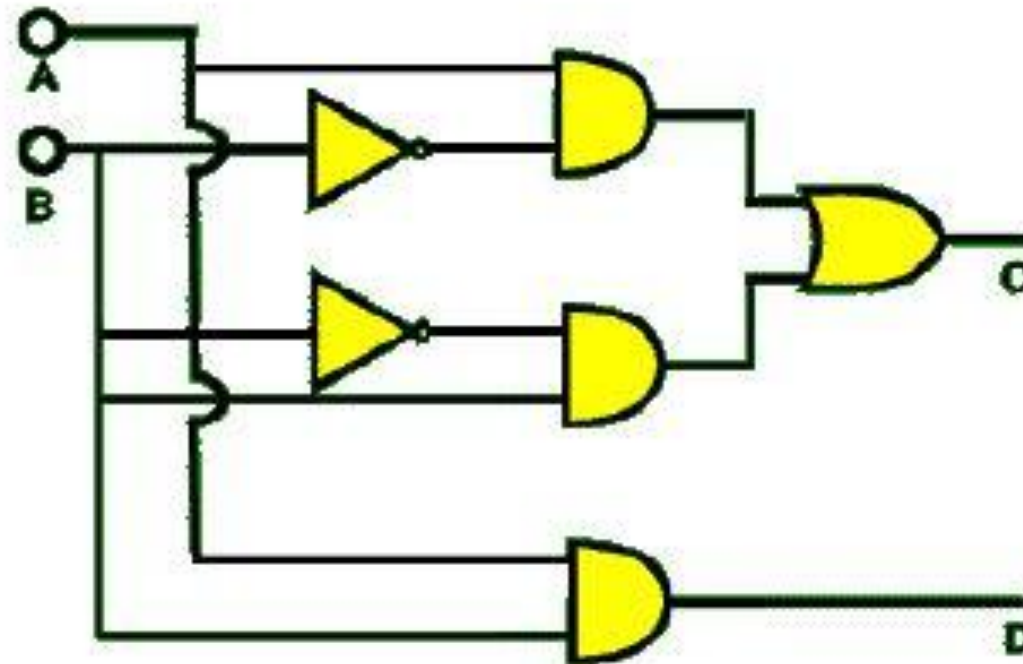
Assembly Language Programming

Result:

S.NO	A	B	C	D	Output (F)
1	0	0	0	0	1
2	0	0	0	1	
3	0	0	1	0	
4	0	0	1	1	
5	0	1	0	0	
6	0	1	0	1	
7	0	1	1	0	
8	0	1	1	1	
9	1	0	0	0	
10	1	0	0	1	
11	1	0	1	0	
12	1	0	1	1	
13	1	1	0	0	
14	1	1	0	1	0
15	1	1	1	0	
16	1	1	1	1	

Assembly Language Programming

Question: To develop an assembly code for 8051 Microcontroller, to implement the given digital circuit using KEIL development tool.



Assembly Language Programming

Solution:

ORG 0000H

SETB ACC.0 ;Input A

SETB ACC.1 ;Input B

MOV C, ACC.1

CPL C

ANL C, ACC.0

MOV ACC.2, C ;Output C

MOV C, ACC.0

ANL C, ACC.1

MOV ACC.3, C ;Output D

H: SJMP H

END

Assembly Language Programming

Result:

S.NO	A	B	Output (C)	Output (D)
1	0	0		
2	0	1		
3	1	0		
4	1	1		

Any Questions?



Thank you!

